BEING A THICKENING OF ALGEBRAIC SPACES IS FPQC LOCAL ON THE TARGET

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-more-morphisms.tex; no statement or proof step was rewritten, reordered or removed. The author's closing sentence omitting the first order case is reproduced as written.

Editorial note. This card is the generalisation that replaces the scheme-level card STK-morph-descend-property-thick (more-morphisms.tex, tag 0BPF), withdrawn under the situation-A criterion; the withdrawn registration sits in operations/new-material-registrations/withdrawn/ and the job is kept.

Editorial note. Preamble provenance, as required for this source: the first line of the pinned chapter file is \input{preamble}, and the native preamble was not staged with the source. The retained preamble is the project's single Stacks-wide preamble (the same 217-line file used by the already published chow and etale-cohomology cards), copied mechanically line for line from sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Not one character was changed, no package was added, no macro was defined, and the excerpt serves only as preamble, never as a span coordinate.

Allowed standard prerequisites (editorial).

- Thickening of algebraic spaces, and the reduction from first order to finite order. Same chapter, the definition of thickening and the discussion in the section on thickenings; the statement itself mentions first order thickenings. Supplies the notion whose local nature is asserted, and the reduction principle the author invokes for the first order case.
- Descent of the property of being a closed immersion for algebraic spaces along an fpqc covering. Descent on Spaces, Lemma labelled spaces-descent-lemma-descending-property-closed-immersion (canonical Stacks tag 0420); invoked at proof lines 1978-1979. Supplies that X -> X' is a closed immersion once its base changes are; the surjectivity step and the assembly into a thickening are retained in the counted span.

Source reference. Original lemma 0420: sources/stacks-chow/tags.txt, source lines 5311–5311. Cross-chapter standard prerequisite invoked once at proof lines 1978-1979 to descend the closed-immersion property for algebraic spaces; the surjectivity step and the conclusion are retained in the counted span.

Lemma 0.1. *The property of being a thickening is fpqc local. Similarly for first order thickenings.*

Proof. The statement means the following: Let S be a scheme and let $X \to X'$ be a morphism of algebraic spaces over S. Let $\{g_i : X'_i \to X'\}$ be an fpqc covering of algebraic spaces such that the base change $X_i \to X'_i$ is a thickening for all i. Then $X \to X'$ is a thickening. Since the morphisms g_i are jointly surjective we conclude

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

that $X \to X'$ is surjective. By Descent on Spaces, Lemma 0420 we conclude that $X \to X'$ is a closed immersion. Thus $X \to X'$ is a thickening. We omit the proof in the case of first order thickenings. $\square$